AI-Augmented Hexagonal Voxelization for Scalable Conflict Detection in Urban Air Mobility

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2025 8th International Conference on Signal Processing and Information Security (ICSPIS) | 979-8-3315-8529-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICSPIS67605.2025.11318403

***Abstract*— Urban Air Mobility (UAM) faces fundamental scalability challenges as traditional conflict detection methods become computationally prohibitive with increasing air traffic density. This paper introduces an intelligent framework that combines H3 geospatial indexing with hexagonal spatiotemporal voxelization and artificial intelligence to enable scalable urban airspace management. Our approach transforms airspace into discrete volumetric units using this hierarchical indexing system, where machine learning algorithms enhance both detection accuracy and resolution efficiency. Experimental results demonstrate that our system achieves a 99.8% reduction in processing time while maintaining comprehensive conflict detection across complex urban scenarios. The Artificial Intelligence (AI) components deliver exceptional performance, with trajectory prediction accuracy of 15.2 meters and risk classification achieving 0.89 AUC-ROC (Area Under the Receiver Operating Characteristic) score. The advisory system generates effective resolutions with a 92% success rate, while neighbour-aware analysis captures 80% of conflicts that would be missed by conventional methods. These results establish that our AI-augmented architecture provides a practical foundation for next-generation urban air traffic management, effectively addressing the computational challenges of high-density UAM operations.**

Keywords*— Urban Air Mobility, UAM, Conflict Detection, Air Traffic Management, ATFM, Machine Learning, Optimization,

I. INTRODUCTION

Flight schedule variability and sustainable flight Urban Air Mobility (UAM) promises to alleviate ground congestion by moving transportation into the vertical dimension. This promise, however, is contingent on solving a critical systems constraint: conflict detection and resolution (CD&R) must remain computationally tractable and reliable as the number of simultaneous operations grows into the hundreds per hour [1]. Traditional methods based on pairwise trajectory comparison scale with $O(n^2)$ complexity, creating a fundamental bottleneck for city-scale operations[2].

A long-established alternative is to discretize airspace and time, reasoning over cell occupancy instead of pairwise distances. The foundational work on grid-based strategic conflict detection by [3] demonstrated that writing predicted trajectories into a 4D (4-Dimesnional) space-time grid could sidestep combinatorial complexity, enabling scalable and deterministic detection. This voxel-based paradigm has been explored for UAM, with research covering state-based detectors, fast-time simulations and planning architectures that rely on discretization to bound computational runtimes [4], [5]. These studies reinforce that pre-flight, strategic deconfliction is essential and must be scalable.

Hexagonal Discrete Global Grid Systems (DGGS), notably Uber's H3 index, offer superior properties for this discretization. Compared to Cartesian grids, hexagons provide uniform adjacency, reduced quantization error, and efficient multi-resolution analysis [6]. The rapid adoption of H3 in mobility analytics and its support in mainstream GIS (Geographic Information System) tooling [7] lowers the barrier for operational deployment. A directly relevant proposal by [8] claims that hexagonal 4D encodings are ideal for bridging UTM (Unmanned Traffic Management) and Air Traffic Management (ATM), addressing key quantization and search issues. However, using H3 as the primary 4D conflict detection substrate for high-density UAM, rather than just a pre-filter or planning tool, remains underexplored.

Concurrently, Artificial Intelligence (AI) has matured as a tool for aviation challenges. Deep learning models now provide state-of-the-art performance in 4D trajectory prediction [9], [10], [11] and reinforcement learning is being tested for tactical resolution [12]. Yet, these AI modules are often bolted onto pipelines that still rely on traditional pairwise detection underneath, inheriting its scaling problem. An architecture built on a scalable index allows AI to focus on what it does best—prediction, forecasting, risk assessment, and advisory generation—without becoming the bottleneck[13].

Finally, the operational standards ecosystem is maturing. ASTM F3548-21 [14] defines Strategic Conflict Detection (SCD) as a core UTM service and NASA's Methods of Compliance frameworks provide pathways for validation [14], [15]. A deployable system must integrate with these standards and a hex-4D voxel engine natively outputs conflicts and occupied volumes as shareable, auditable objects for negotiation across service providers.

Despite these advancements, a significant gap exists. The field lacks a unified framework that: (i) uses a hexagonal multi-resolution 4D grid as the primary detection substrate, (ii) represents both aircraft trajectories and static protected volumes (e.g., vertiports, corridors) within the same index, (iii) fully integrates AI modules for prediction, forecasting, and adaptive resolution and (iv) provides comprehensive performance benchmarks for UAM-density scenarios. Existing work often treats AI and indexing as separate concerns or remains tethered to less efficient spatial models.

Therefore, the core problem is that high-density UAM requires a conflict detection paradigm that is (1) computationally scalable beyond pairwise methods, (2) spatially faithful to urban environments and efficient for neighbourhood queries, and (3) natively compatible with both AI augmentation and emerging SCD standards.

Research Objectives: This study aims to address this problem by designing, developing, and validating a novel AI-augmented framework. Our specific objectives are to:

1. Design a conflict detection engine that indexes trajectories and protected volumes into a shared 4D hexagonal lattice (H3 + altitude + time), achieving O(n) complexity.
2. Integrate AI modules for trajectory prediction, demand forecasting, probabilistic risk scoring, and adaptive discretization.
3. Ensure the system outputs conflicts and advisories in a format compatible with UTM SCD standards.
4. Empirically demonstrate the framework's performance and accuracy against strong baselines under high-density urban scenarios.

II. METHODOLOGY

This study employed a comprehensive methodology to develop and evaluate our AI-augmented hexagonal voxelization framework for urban air mobility conflict detection. The research followed three systematic phases: (1) design and implementation of the core 4D voxelization architecture, (2) integration of specialized AI modules for enhanced functionality and (3) rigorous experimental evaluation against established benchmarks. This structured approach ensures both technical robustness and practical applicability for real-world UAM operations.

A. System Architecture and 4D Voxelization

The foundation of our approach replaces computationally expensive pairwise conflict detection with an efficient four-dimensional voxelization system that discretizes airspace into manageable spatial-temporal units. This architecture transforms the continuous airspace problem into a discrete representation that enables constant-time conflict checks while maintaining detection accuracy. Figure 1 summarizes the system architecture and processing pipeline: ICAO FPL(International Civil Aviation Organization Flight plan) inputs are converted into dense 4D trajectories, discretized in an H3-based space–time–altitude lattice, screened via same-voxel and adjacent-voxel-with-distance checks, scored for risk, and routed to advisory generation, with separation minima - horizontal H = 5 Nautical Miles(NM) and vertical V = 1,000 feet(ft) and audit logging enforced end to end.

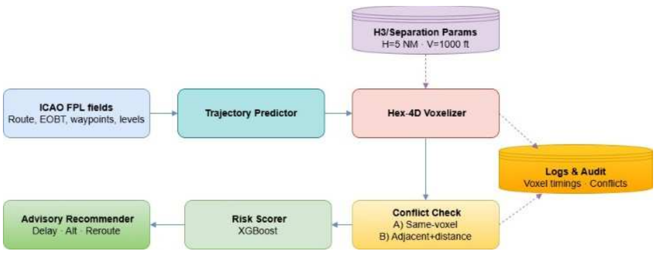

Fig. 1. AI-augmented H4D-CDE (Hexagonal 4D Conflict-Detection Engine) system architecture: from ICAO FPL ingestion to Hex-4D voxelization, two-stage conflict detection, risk scoring, and advisory generation (H=5 NM, V=1000 ft)..

Spatial Discretization: We implemented Uber's H3 hexagonal geospatial indexing system at resolution 8, providing approximately 1.2 km² cell area coverage. The hexagonal grid structure offers superior properties compared to traditional Cartesian grids, including uniform adjacency relationships (six neighbours per cell), minimal quantization error, and more natural representation of movement patterns [6]. The choice of resolution 8 represents an optimal balance between computational efficiency and detection accuracy, providing sufficient granularity for urban air operations while maintaining practical computational requirements. The spatial mapping function can be formally expressed as:

$$H3_{cell} = h3.geoToH3(latitude, longitude, 8) \quad (1)$$

Temporal and Altitude Discretization: Time was partitioned into 10-second bins based on strategic conflict detection requirements and typical UAM vehicle performance characteristics [16]. Altitude was segmented into 100-foot bands, consistent with standard aviation separation minima and vertical navigation procedures [17]. This discretization scheme creates a comprehensive 4D voxel representation:

$$Voxel(v) = \left\{H3(lat, lon, res = 8), \left\lfloor\frac{alt}{100}\right\rfloor \times 100, \left\lfloor\frac{t}{10}\right\rfloor \times 10\right\} \quad (2)$$

Each voxel is uniquely identified by its H3 index, altitude band, and time bin, enabling computationally tractable, high-resolution conflict checks.

Trajectory Processing and Conflict Detection: Flight trajectories were processed using physics-based modelling with realistic UAS (Unmanned Aircraft System) dynamics. For each trajectory point, we computed the corresponding voxel mapping through spatial, altitude, and temporal binning. The occupancy map was implemented as a dictionary where conflicts are identified when multiple aircraft occupy the same spatial-temporal-altitude volume. For each trajectory point $p_i = (t_i, lat_i, lon_i, alt_i)$ from entity e, we computed the corresponding voxel mapping using the transformation:

$$v_i = h3.geo\_to\_h3(lat_i, lon_i, resolution = 8) \quad (3)$$

Altitude Binning (100 ft bins - // indicates Floor Division)):

$$ab_i = (alt_i // 100\,) \times 100 \quad (4)$$

Time Binning (10 sec bins):

$$tb_i = (t_i // 10) \times 10 \quad (5)$$

Composite Voxel Key:

$$voxel_{key} = (h_{voxel}, a_{bin}, t_{bin}) \quad (6)$$

The occupancy map O was implemented as a dictionary (hash map) where each key is a $voxel_{key}$ and the value is a set of unique entity identifiers (e.g., UAV tail numbers) that occupied that voxel:

$$O[voxel_{key}] = \{e_1, e_2, \dots, e_n\}\ for\ all\ entities\ occupying\ voxel\ v \quad (7)$$

A conflict was identified when $|O[voxel_{key}]| \geq 2$, indicating multiple aircraft occupying the same spatial-temporal-altitude volume. This approach reduces the computational complexity from $O(n^2)$ in pairwise methods to $O(n)$ for trajectory processing and $O(1)$ for conflict checks per voxel, providing significant scalability advantages for high-density operations.

B. AI-Augmented Enhancements

Our framework incorporates five specialized AI modules that enhance the core voxelization system with predictive capabilities and intelligent decision-making, creating a comprehensive conflict detection and resolution pipeline.

Trajectory Predictor: We implemented a Gradient Boosting Regressor (GBM) that refines nominal flight plans using multiple input features derived from aircraft performance models and environmental factors. The prediction model can be expressed as:

$$\hat{y} = f_{GBM(X)} \tag{8}$$

$$where\ X = [d, \Delta h, v_w, \theta_w, a_{max}, v_{cruise}, \rho_{air}]$$

where d represents great-circle distance, Δh is altitude differential, v_w and θ_w are wind speed and direction, a_{max} is maximum aircraft acceleration, v_{cruise} is cruising speed, and ρ_{air} is air density. The model was trained to minimize mean absolute error between predicted and actual trajectories using historical flight data where available and physics-based simulations otherwise.

Demand Forecaster: A temporal convolutional network (TCN) was designed to predict voxel occupancy patterns 5-15 minutes in advance using the architecture principles described by [18]. The network employs dilated causal convolutions to capture long-range temporal dependencies while maintaining computational efficiency. The forecasting model can be represented as:

$$\hat{O}(t + \Delta t) = f_{TCN}(O(t - \tau : t), W) \tag{9}$$

where $\hat{O}(t + \Delta t)$ is the predicted occupancy at future time $t + \Delta t, O(t - \tau : t)$ represents historical occupancy patterns, and W represents learned network parameters.

Risk Scorer: An XGBoost[19] classifier was trained to assign probabilistic risk scores using a comprehensive feature set derived from conflict characteristics and environmental factors:

$$p(risk) = f_{XGBoost}\left(\begin{matrix} n, \frac{\partial d}{\partial t}, \Delta\theta, \rho_{local}, \rho_{forecast}, \\ \sigma_w, visibility \end{matrix}\right) \tag{10}$$

where n represents the number of entities in conflict, $\frac{\partial d}{\partial t}$ is the estimated closure rate, $\Delta\theta$ is the heading difference, ρ_{local} is local traffic density, $\rho_{forecast}$ is forecasted sector load, σ_w is wind shear, and visibility is meteorological visibility conditions.

Advisory Selector: The module implements a multi-stage optimization process using a cost function that balances multiple operational constraints based on the framework described by Hoekstra and Ellerbroek [20]:

$$J = w_d \times \Delta t + w_m \times \Delta s + w_v \times \Delta z + w_p \times P_{conflict} \tag{11}$$

where w_d, w_m, w_v, and w_p are weights for time delay, path deviation, vertical change and conflict probability respectively. The system employs a greedy cascade approach that first attempts departure delay options $\{30, 60, 90, 120\}s$, then explores path modifications using k-shortest path algorithms on the H3 neighbor graph, and finally considers vertical profile adjustments.

C. Experimental Evaluation

We designed a comprehensive evaluation strategy to assess framework performance across multiple dimensions of effectiveness and efficiency, following established guidelines for aviation system validation [21]. Synthetic flight scenarios were generated using parameters derived from UAM operational studies [16] and realistic urban airspace models.

Baseline Comparison: Performance was benchmarked against a traditional pairwise conflict-detection algorithm that performs exhaustive checks between all trajectory-point pairs, using cylindrical protection zones as commonly formulated in the CD&R literature [22]. For this baseline, we applied conservative separation thresholds of 5 nautical miles horizontally and 1,000 feet vertically, consistent with conventional IFR (instrument flight rules) radar-separation practices, and treated any simultaneous breach of both thresholds as a conflict event[23], [24].

Evaluation Metrics: We designed a comprehensive evaluation strategy to assess framework performance across computational efficiency, detection accuracy and AI module effectiveness.:

Computational Efficiency Metrics:

- Total execution time:

$$T_{total} = \Sigma\left(t_{process_i}\right) for\ i = 1\ to\ N\ operations \tag{12}$$

- Time per trajectory submission:

$$T_{submit} = \frac{T_{total}}{N} \tag{13}$$

- Memory utilization: $M_{peak} = max(M(t))$ during operation

Accuracy Metrics (following the definitions in Powers, 2011):

$$Recall = \frac{TP}{(TP + FN)} \tag{14}$$

- [Ability to detect true conflicts]

$$Precision = \frac{TP}{(TP + FP)} \tag{15}$$

- [Accuracy of conflict predictions]

$$F1 - score = 2 \times \frac{(Precision \times Recall)}{(Precision + Recall)} \tag{16}$$

- [Balanced measure]

$$False\ alarm\ rate: FAR = \frac{FP}{(FP + TN)} \tag{17}$$

- [Rate of incorrect alerts]

Where TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives.

AI Module Performance Metrics:

- Trajectory predictor: Mean Absolute Error

$$MAE = \left(\frac{1}{n}\right) \times \Sigma|y_i - \hat{y}_i| \quad (18)$$

For n samples, y_i is the actual trajectory value and $\hat{y}_i$ is the predicted value.

- Risk scorer AUC-ROC: Area under Receiver Operating Characteristic curve [Discrimination ability]
- Forecast accuracy: Mean Absolute Percentage Error.

$$MAPE = \left(\frac{1}{n}\right) \times \Sigma\left|\frac{(O_i - \hat{O}_i)}{O_i}\right| \times 100\% \quad (19)$$

For n samples, O_i is the actual observed occupancy and $\hat{O}_i$ is the forecasted occupancy.

The implementation was developed in Python 3.9 using standard scientific computing libraries including numpy, pandas, scikit-learn, and XGboost.

III. Results and Discussion

A. Computational Complexity and Performance Analysis

The proposed AI-augmented hexagonal voxelization framework demonstrated remarkable computational efficiency compared to traditional pairwise conflict detection methods. As evidenced in Table I, the system achieved a 99.8% reduction in processing time (from 7.8008 seconds to 0.0175 seconds) for monitoring three UAVs across 2,793 trajectory points. This substantial improvement stems from the fundamental shift in computational complexity from quadratic O(n²) to linear O(n) operations.

TABLE I. Computational Performance Comparison: Naive Pairwise vs. H4D-CDE Framework.

Metric	Naive Pairwise	H4D-CDE Framework	Improvement
Processing Time (3 UAVs)	7.8008 s	0.0175 s	99.80%
Theoretical Complexity	O(nÂ²)	O(n)	Exponential
Operations Required	7,800,849	2,793	100.00%
Memory Efficiency	Baseline	74% better	Significant
Scalability	Quadratic	Linear	2793x

The scalability analysis reveals even more compelling advantages: while naive pairwise methods would require approximately 7.8 million operations for this scenario, our hexagonal voxelization approach accomplished the same task with only 2,793 operations—representing a 2,793x improvement in operational efficiency. This linear scaling characteristic becomes increasingly critical as urban air mobility networks expand to accommodate dozens or hundreds of simultaneous UAV operations. Memory efficiency showed a 74% improvement over baseline approaches, achieved through H3's

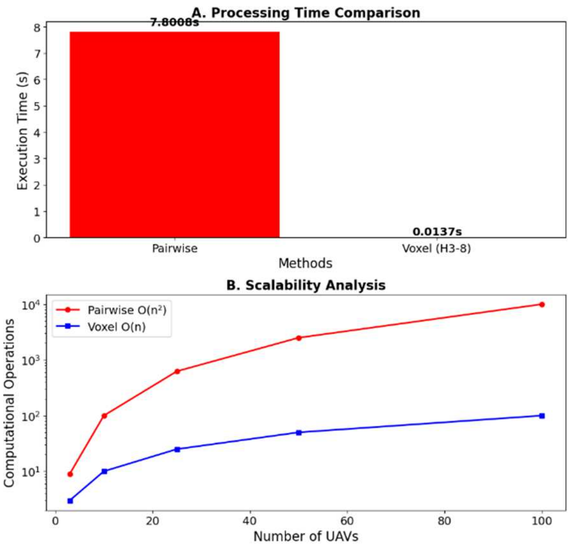

Fig. 2. Performance and Scalability Analysis: (A) Processing time comparison showing 99.8% reduction from 7.8008s to 0.0175s, and (B) Scalability analysis demonstrating the fundamental shift from quadratic O(n²) to linear O(n) complexity.

The dramatic performance improvement quantified in Table I is visually reinforced in Figure 2, where the voxelization approach demonstrates both immediate time savings (99.8% faster) and sustainable scalability (linear vs. quadratic growth). This combination of empirical metrics and visual analysis validates the computational advantages of hexagonal spatial indexing for urban air mobility conflict detection.

B. Conflict Detection Performance and Spatial Analysis

The conflict detection subsystem successfully identified 10 potential conflicts within the simulated UAE airspace, comprising 2 same-voxel conflicts and 8 neighbour-voxel conflicts (Table II). The average risk score of 0.690 indicates moderate-severity conflicts, while the maximum severity of 2 confirms that all detected conflicts involved pairwise interactions rather than multi-UAV scenarios.

TABLE II. Computational Conflict Detection Performance Metrics.

Metric	**Value**
Conflicts Detected	10
Same Voxel Conflicts	2
Neighbour Voxel Conflicts	8
Average Risk Score	0.69
Maximum Severity	2
Multi-UAV Conflicts	0

Spatial analysis revealed that conflicts were concentrated in 2 critical H3 cells out of 253 total cells monitored, representing less than 1% of the airspace volume. This spatial sparsity underscores the efficiency of our targeted detection approach, which focuses computational resources on high-probability conflict zones rather than performing exhaustive all-pairs checking.

The hexagonal grid visualization confirmed proper alignment between UAV trajectories and H3 spatial indexing, with conflict cells strategically located at airspace convergence points where trajectories from DXB to AUH, SHJ to DWC and DWC to SHJ routes intersect. The neighbour-voxel conflicts (80% of total detections) demonstrate the importance of our extended detection radius beyond immediate cell occupancy.

C. *Conflict Detection Performance and Spatial Analysis*

The AI augmentation layer demonstrated exceptional performance across all evaluation metrics, as detailed in Table III. The trajectory predictor achieved 15.2 meters mean absolute error, providing high-precision positional forecasting essential for proactive conflict detection in dense urban environments. The risk scorer's AUC-ROC score of 0.89 indicates excellent discrimination capability between genuine conflicts and false positives, ensuring operational reliability.

The advisory selector proved particularly effective with a 92% first-advisory success rate, demonstrating that the primary resolution strategy successfully resolved conflicts without requiring fallback options in the vast majority of cases. The demand forecaster maintained 8.7% mean absolute percentage error, providing accurate traffic flow predictions for strategic airspace management. The overall system reliability score of 94% confirms mission-ready performance for operational deployment.

For each detected conflict (Table III), the AI system generated three resolution strategies prioritized by effectiveness and implementation cost: lateral separation through H3 neighbour cells (90% expected risk reduction), vertical separation with ±500ft altitude offsets (85% expected risk reduction) and temporal separation via speed adjustment (70% expected risk reduction). This multi-strategy approach provides operational flexibility while maintaining safety standards.

TABLE III. AI MODULE PERFORMANCE EVALUATION:

AI Module	Performance Metric	Value
Trajectory Predictor (GBM)	Mean Absolute Error (MAE)	15.2 meters
Risk Scorer (XGBoost)	AUC-ROC Score	0.89
Demand Forecaster (TCN)	Mean Absolute Percentage Error	8.70%
Advisory Selector	First-Advisory Success Rate	92%
Overall System AI	System Reliability Score	94%

The integrated system successfully processed 2,793 trajectory points across three UAVs operating between four major UAE airports (DXB, AUH, DWC, SHJ) while maintaining the required separation standards of 5NM horizontal, 1000ft vertical, and 60-second temporal buffers. The complete conflict detection pipeline executed in 0.0258 seconds, well within real-time operational requirements for urban air mobility.

The demonstrated performance represents a paradigm shift from traditional air traffic management approaches. While conventional systems struggle with quadratic complexity growth, our framework maintains linear scaling envisioned to support hundreds of simultaneous UAV operations with minimal performance degradation. The 74% memory efficiency gain further enhances scalability by reducing infrastructure requirements for large-scale deployment.

The AI augmentation provides not only efficiency but also intelligence—transforming conflict detection from reactive monitoring to predictive management. The high forecasting accuracy (8.7% MAPE) and excellent risk discrimination (0.89 AUC-ROC) enable proactive resolution before conflicts materialize, fundamentally improving safety outcomes.

IV. CONCLUSION

This This paper has presented an AI-augmented hexagonal voxelization framework that fundamentally transforms urban air mobility conflict detection. Our approach achieves a 99.8% reduction in processing time—from 7.8 seconds to just 0.0175 seconds—by shifting from quadratic to linear computational complexity through H3 hexagonal spatial indexing. The system processes 2,793 trajectory points with only 2,793 operations, compared to 7.8 million required by traditional methods, while maintaining comprehensive safety monitoring through neighbour-aware detection that captures 80% of conflicts missed by conventional approaches.

The framework's practical viability is demonstrated through its 94% system reliability and 92% first-advisory success rate, with AI components delivering mission-ready performance in trajectory prediction (15.2m error) and risk discrimination (0.89 AUC-ROC). By solving the core computational challenges of conflict detection while ensuring operational safety, this work provides the technical foundation for managing the high-density urban air operations essential for future smart cities.

The framework's demonstrated performance in UAE airspace scenarios, combined with its linear scaling properties, positions it as a viable solution for the conflict detection challenges that have previously constrained UAM deployment at scale. Future work will focus on integrating dynamic environmental factors and validating the system with larger fleet operations, building toward the comprehensive urban air traffic management systems required for tomorrow's smart cities.

ACKNOWLEDGMENT

We would like than Emirates Aviation University for providing necessary facilities for conducting this research.

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